



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS: IX
SUBJECT: BIOLOGY

FULL MARKS: 40
DATE: 31.10.22

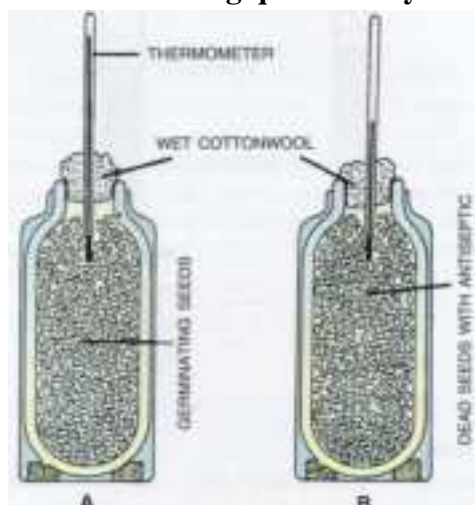
General Instructions:

- This paper consists of two printed pages.
- Answers should be to the point.
- Copy the question number carefully before answering the questions.
- Marks will be deducted for spelling errors.

1. Choose the correct answer: [5]
- a. The fate of ovary after fertilization
i. Testa, ii. Fruit, iii. Pericarp, iv. Seed
 - b. Monocot Albuminous Seed
i. Poppy, ii. Cereal, iii. Custard Apple, iv. *Vallisneria*.
 - c. Glycolysis is a process in which
i. Glycogen is converted to glucose,
ii. Glucose is converted into pyruvate,
iii. Glucose is converted to carbon dioxide and water
iv. Glucose is converted to Ethanol
 - d. Exine is the outer wall of
i. Ovule, ii. Pollen grain, iii. Stigma, iv. Ovary
 - e. The process by which anthers are removed from young flowers are
i. Bagging, ii. Hybridization, iii. Pollination, iv. Emasculation
2. Name the following: [5]
- a. When the maturation time of the reproductive parts of a flower are same.
 - b. A chemical which absorbs oxygen.
 - c. A distinct whitish oval scar on the concave side of the seed.
 - d. Full form of ATP.
 - e. When the stigma of a flower matures early than the anther.
3. Choose the odd one out and write the correct category to which the others belong: [5]
- Example- Large Intestine, Stomach, Mouth, Brain.
- Odd one- Brain, Category- Organs of Digestive System
- i. Zygote, Autogamy, Allogamy, Geitonogamy.
 - ii. *Rhizophora*, Maize, Pea, Gram.

- iii. Glucose, Carbon dioxide, Water vapour, 38ATP.
- iv. Lady Finger, Sweet Pea, *Salvia*, Custard Apple.
- v. Ovary, Cotyledon, Hypocotyl, Epicotyl.

4. Differentiate between the following according to the criterion given in brackets: [5]
- i. Anaerobic respiration in plants and Aerobic respiration (Energy liberated)
 - ii. Bean and Maize (Number of cotyledons)
 - iii. Testa and Tegmen (Definition)
 - iv. Autogamy and Allogamy (Definition)
 - v. Self-sterility and Unisexuality (Definition)
5. i. Define hypogeal germination. [1]
- ii. "Seeds sown deep into the soil fail to germinate". Mention two reasons to explain this statement. [2]
 - iii. Mention two disadvantages of cross pollination. [2]
 - iv. Briefly explain double fertilization in two distinct points. [2]
 - v. Draw a neat labelled diagram of L.S. of a Dicot Seed. [3]
6. i. Define Aerobic Respiration. [1]
- ii. Give reasons for the following: [2]
 - a. Antiseptic is used on the dead seeds during the experiments of respiration.
 - b. KOH is used in the experiments of respiration.
 - iii. Explain the importance of water during germination. (Two points) [2]
 - iv. Mention the protective sheath which encloses the plumule. Why is maize known as a grain and not a seed? [2]
- v. Answer the following questions by observing the diagram given below: [3]



- a. Mention the aim of the above experiment.
- b. Mention with reasons what will you observe in the thermometers of Flask A and Flask B after sometime?